Week : 9
الأسبوع 9: من - إلى

Subject/:الموضوع Chemistry (IB DP) **Unit Title : عنوان الوحدة Structure 1.4 – The mole** **Grade: 11 (DP1)**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 **S1.4.5 Molar concentration**	**Learning target (DOK 2):** I can explain how molar concentration links the amount of solute to the volume of solution, and use [ ], n = CV and conversions between g dm^{-3} and mol dm^{-3} to solve problems.	**Formative:** warm-up on mini whiteboards, hinge question, practice tasks on n = CV and unit conversions, and the exit ticket. **Differentiation:** step-by-step worked examples for Must; mixed-unit problems for Should; serial dilution and calibration-curve challenge for Could. **UDL strategies:** 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Interactive: PhET Molarity simulation – see how changing the amount of solute and the volume changes concentration. Class Teams (Week 9 folder): lesson slides, n = CV practice questions and worked solutions. IB data booklet: n = CV is given.
Lesson 2 **الحصة الثانية** **S1.4.6 Avogadro's law**	**Learning target (DOK 2):** I can explain Avogadro's law and use it to relate the mole ratio of reactants and products to the volumes of gases measured at the same temperature and pressure.	**Formative:** retrieval starter, mini-whiteboard checks on gas-volume mole ratios, 'spot the mistake' and a Must/Should/Could exit ticket. **Differentiation:** choose-your-challenge practice with self-marking; extension question on when real gases deviate from ideal behaviour (Structure 1.5). **UDL strategies:** 2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Interactive: PhET Gas Properties simulation – see how the number of gas particles and the volume are related at constant temperature and pressure. Class Teams (Week 9 folder): lesson slides and gas-volume practice questions.

Lesson 3 **الحصة الثالثة** **Grow Day: summative feedback**	**Learning target (DOK 2):** I can use feedback from the summative to identify the types of errors I made in Structure 1.4 problems and correct them with justified working.	**Formative:** personal error analysis of the summative (errors sorted by type), targeted feedback conversations, corrected solutions and an exit ticket. **Differentiation:** targeted practice stations chosen from summative results: reteach for Must; mixed problems for Should; challenge problems for Could. **UDL strategies:** 9.3 Develop self-assessment and self-reflection with tools, charts, and feedback displays. 3.3 Guide processing with explicit steps, chunking, and interactive models. 8.2 Vary challenge level with differentiated tools, resources, and difficulty.	Class Teams: summative feedback, corrected-solutions task and revision resources for Structure 1.4. At home: ask your child to explain their two most common errors and how they corrected them.
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**	Hello Parents, Please encourage your child to review and practice Science regularly each week using the resources available in their class Microsoft Teams - class material folder which is organized by week. The presentations, readings, videos, practice activities, etc. posted in Teams are the primary learning materials and should be used throughout the unit to support understanding and achievement. Science study guides will provide basic assessment information, topics, and expectations; however, they are intended as a guide only and will not contain detailed study materials, these are found in the class teams. Effective preparation for graded formatives and summatives requires students to actively engage with the learning materials provided in their class teams. Thank you for partnering with us to support your child's learning and success. Ms. Stacey (Science Coordinator)		

Week 9 الأسبوع 9

Technology Integration دمج التكنولوجيا	Videos posted on Teams; PhET Molarity and Gas Properties simulations; mini whiteboards; scientific calculators; digital IB data booklet; slides shared after each lesson.
Cross-curricular links الروابط بين المواد الدراسية	**Mathematics:** rearranging equations, unit conversions (cm^3, dm^3) and significant figures. **UAE Social Studies:** water quality testing and desalination brine as real contexts. **Physics:** SI units; gas volume, temperature and pressure.
External Assessment Integration دمج التقييمات الخارجية	IB DP Chemistry Papers 1 and 2: data-booklet calculations (n = CV), gas-volume questions and multi-step structured questions using the command terms calculate, determine, deduce and justify.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	**Thinking:** justifying answers and spotting errors. **Self-management:** using summative feedback to set targets and rating themselves 1–5 against the success criteria. **Social:** pair work and peer feedback; choose-your-challenge tasks offer stretch without pressure.
Standards المعايير	• **S1.4.5** The molar concentration is determined by the amount of solute and the volume of solution. • **S1.4.6** Avogadro's law states that equal volumes of all gases measured under the same conditions of temperature and pressure contain equal numbers of molecules.
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	**Guiding question:** How do we quantify matter in solutions and in gases? **Tool 1:** Choice of glassware for a standard solution and a serial dilution; using a calibration curve to find a concentration. **Structure 1.5:** Avogadro's law applies to ideal gases. Under what conditions might the behaviour of a real gas deviate most from an ideal gas? **Lesson 3 (Grow Day):** How can analysing our mistakes help us learn more effectively?
Success Criteria معايير النجاح **(Measurable)**	● **Must (DOK 1–2):** use n = CV to calculate amount, volume or concentration, and state Avogadro's law. ■ **Should (DOK 2–3):** convert between g dm^{-3} and mol dm^{-3} and use mole ratios with gas volumes to solve problems. ▲ **Could (DOK 3–4):** justify multi-step problems and evaluate when a real gas deviates most from ideal behaviour. **Evidence:** hinge questions, exit tickets and the Grow Day error analysis.
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء — Global Understanding التفاهم العالمي — Heritage التراث المجال Area: Innovation and Infrastructure التطبيق Implementation: Students apply molar concentration and gas-volume ideas to UAE contexts: preparing standard solutions for water-quality testing, the concentration of brine from desalination, and gas volumes in methane combustion.